
TOWARD REALISTIC BREAST SHAPES IN A FINITE-ELEMENT MODEL FOR CANCER SCREENING

Internship at Sioux Technologies
16-02-2026 to 22-05-2026

Femke Storm
1336819

Sioux Technologies Supervisors
dr. Timo van Opstal
dr. Aybuke Kazaz
dr. Wouter Wilson

TU/e Supervisor
dr. Federica Eduati

Final report

Eindhoven, 26-05-2026



Department of Biomedical Engineering
Computational Biology

Abstract

In an effort to reduce the discomfort, cost and challenges associated with mammography for breast cancer screening, the Early Warning Scan (EWS) aims to develop a noninvasive breast cancer screening method using a camera-based setup. The method uses the dynamic behavior of the breast as an input for an AI algorithm to predict the likelihood of a breast tumor, and assess the need for further medical evaluation. To support the development of the EWS, Sioux Technologies developed a finite-element method (FEM) model of the breast with the goal of generating a low-cost training data set with and without tumors. This project focused on the geometry of the static state of the breast under gravity. Strides were made in improving the geometry of the breast to include more realistic asymmetric properties and to further capture the variation of breast shape observed in the population. Furthermore, the Holzapfel-Gasser-Ogden material model was implemented to include the anisotropic behavior of the breast due to the Cooper Ligaments. The results of these geometric and material changes were evaluated using an algorithm to optimize the model parameters to match the model-generated breast shapes to an available dataset of 3D-scanned breasts. While the changes improve the realism of the resulting breast shape, challenges remain in accurately capturing the softness of breast tissue.

Keywords: breast cancer, cancer screening, finite-element method, breast geometry, anisotropic tissue modeling, geometry optimization

Contents

1	Introduction	3
2	Background	4
	2.1 Anatomy of the breast	4
	2.2 Finite-Element Method	4
	2.3 Previous work	6
3	Methods	7
	3.1 Breast geometry	7
	3.2 Meshing	7
	3.3 Finite Element Model	8
	3.4 Comparison of the model with real breast shapes	12
4	Results	15
	4.1 Variable geometry	15
	4.2 Suspensory ligaments	16
	4.3 Parameter optimization	17
5	Discussion	19
	5.1 Breast geometry	19
	5.2 Material parameters	19
	5.3 Element inversion	19
	5.4 Suspensory Ligaments	20
	5.5 Parameter optimization	20
	5.6 Summary	21

1 Introduction

Breast cancer is one of the most prevalent cancer types worldwide[1]. In highly developed countries, 1 in 12 women will be diagnosed with breast cancer in their lifetime , where approximately half of breast cancer cases have no identifiable risk factor [2].

Breast cancer starts with mutated cells in the glandular tissue, where the milk glands and ducts reside, and later spreads to the rest of the breast and the surrounding lymph nodes. Oftentimes, the first noticed symptom is a lump that can be felt by hand. However, breast cancer has several other lesser-known symptoms, such as an inverted nipple, dimpled or discolored skin or increased breast stiffness [3]. Unfortunately, these symptoms often go unnoticed, and the disease only gets found out when it has progressed.

One key predictor of survival is the stage at which the disease is diagnosed, where the disease has up to 100% 5-year survival rate for women in the US diagnosed at stage I. Diagnoses at stage IV see a reduced 5-year survival rate of 86%. [4]. In most developed countries, extensive screening programmes for women above age 40 are set up to increase the early detection rates. The most frequent are self-exams and mammography. Up to 60% of diagnoses of breast cancer are as a result of routine mammography [5, 6], leading to earlier detection of the disease and therefore better global survival rates. However, routine mammography has its disadvantages; frequent exposure to x-ray radiation poses a risk of DNA damage [7], dense breast tissue in (often young) women makes tumor detection difficult [8] and some women choose to opt out of screening due to fear of the procedure [9].

The Early Warning Scan (EWS) aims to develop a novel method for breast cancer screening using noninvasive measurements [10]. The concept involves a 52-camera setup which records movement, temperate, and coloration of the breast, which is then analyzed by an AI model to predict an increased chance of tumor presence in the breast. The development of such an AI model requires a large amount of training data, which is difficult and expensive to obtain.

Therefore, Sioux Technologies supports the development of the EWS method by providing a method to generate large amounts of realistic synthetic training data of the movement of the breast both with and without tumors present. The hypothesis is that tumor tissue, being much stiffer than the surrounding tissues, alters the dynamic behavior of the breast in a way that can be detected externally. To simulate this, a Finite-Element Method (FEM) model can be used to accurately model the dynamic behavior of the breast in response to gravity and other motion. A simple version of the FEM model was previously established, and this project improves upon the existing model.

This project consists of three objectives. The first is improving the geometry of the model to better include asymmetric properties of the breast. The second objective is implementing a new material model in the adipose tissue to model the presence of an important structural feature of the breast. The third objective is to validate the model by comparing its simulated breast shapes to a dataset of real-life 3D imaging. Together, these objectives aim to increase the realism of the generated breast shapes, to work towards the generation of accurate and reliable simulations.

2 Background

2.1 Anatomy of the breast

In female adults, the breast (also called *mammary gland*) is a system of lobules and ducts surrounded by adipose tissue, lymph nodes and connective tissue. The lobules can produce milk, which is transported through the ducts towards the nipple, situated at the front of the breast. The breast does not contain muscle, but is situated on the chest wall muscles (*pectoralis major and minor*). The breast is covered with skin, which is connected to the breast tissue with a layer of connective tissue as well as with suspensory ligaments. [11, 12]

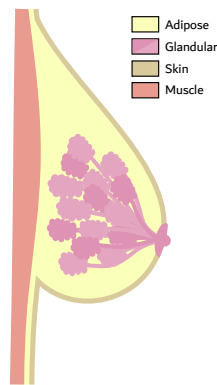


Figure 1: Simplified depiction of the anatomy of the breast. (Loosely based on Jesinger [12])

2.1.1 Glands

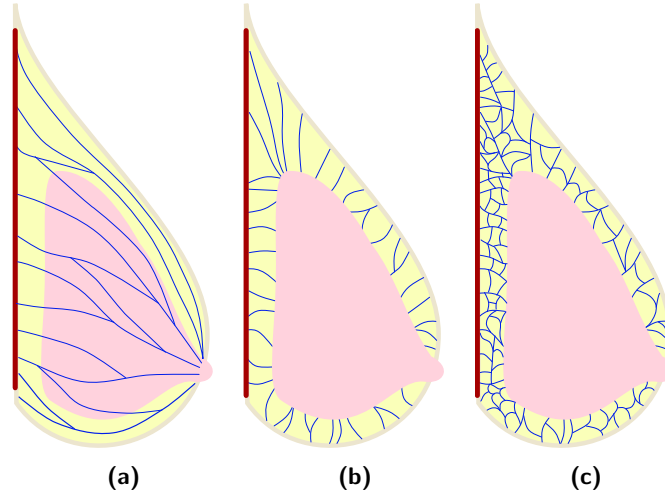
The mammary system in the breast is composed of 15-20 lobules [11], and each lobule is connected to the nipple with a duct. The size of the lobules can vary greatly throughout the lifetime of a woman, with the most major changes appearing before, during and after breastfeeding, but more minor changes in glandular volume also appear throughout the menstrual cycle [11, 13].

2.1.2 Cooper's Ligaments

The suspensory ligaments, or Cooper's ligaments, are thin ligaments situated in the subcutaneous adipose layer of the breast. They are thought to be largely responsible for the shape of the breast [14]. Descriptions of the exact position, direction and structure of the suspensory ligaments vary throughout literature, where the ligament are sometimes described as thin wall-like structures connecting the glandular tissue to the skin (Fig 2a, 2b) [15], while others describe multiple layers of adipose-filled pockets of ligament between the skin and the glandular tissue (Fig 2c) [14, 16].

2.2 Finite-Element Method

The Finite-Element Method (FEM) is a numerical method to provide approximations of the solution for differential equations in a continuum by dividing it into smaller elements and approximating the solution for these elements. These partial solutions are then used to compute an approximation for



the global solution of the continuum. In this project, the feature of interest is the displacement of the tissue under a load. Therefore it is important to distinguish between the initial configuration and the current configuration. A point initially at position $\mathbf{X}$ can move to a new position $\mathbf{x}$. These displacements define the deformation map of the system. The local stretch, rotation and distortion can be described by deformation gradient tensor $\mathbf{F}$.

$$\mathbf{F} = \frac{\delta \mathbf{x}}{\delta \mathbf{X}} \quad (1)$$

From this deformation gradient, a measure of the strain independent from large body rotation can be computed using the right Cauchy-Green deformation tensor $\mathbf{C}$

$$\mathbf{C} = \mathbf{F}^T \mathbf{F}. \quad (2)$$

Hyperelastic material models, which are often used to describe biological materials, are usually expressed in scalars called strain invariants. These invariants (typically I_1 , I_2 , I_3) are derived from $\mathbf{C}$ and provide a way to describe the deformation without referring to a coordinate system.

$$I_1 = \text{tr}(\mathbf{C}) \quad (3)$$

$$I_2 = \frac{1}{2} [(\text{tr}(\mathbf{C}))^2 - \text{tr}(\mathbf{C}^2)] \quad (4)$$

$$I_3 = \det(\mathbf{C}) \quad (5)$$

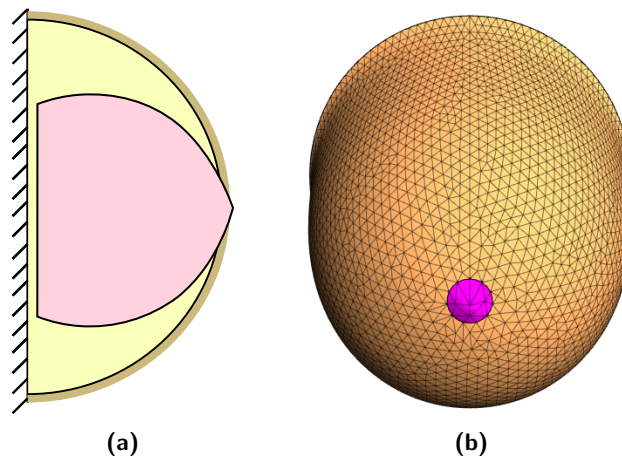
Material models are described using a strain-energy function Ψ . Biological tissues mostly consisting of water are often considered incompressible. Therefore, an uncoupled model can be used, which splits Ψ into a deviatoric part $\tilde{\Psi}$, describing the deformation of the element, and a volumetric part $U(J)$ describing volume change. In FEM, this strain-energy function is used to compute the internal energy of the system over all elements. The displacement field is then computed by minimizing the total internal energy, taking into account boundary conditions and applied loads, leading to the equilibrium state of the system.

2.3 Previous work

Previously, a FEM model of the breast was developed, which models the breast as a half sphere (see Fig. 3a with gravity load. This results in the deformed breast in Figure 3b).

A pipeline was built to generate the FEM model of the breast based on a number of parameters (e.g. radius of the breast, material parameters), deform it under gravity, and perform a parabolic jump on it to investigate the dynamic behavior of the breast.

This modeled included the option to model a tumor in the breast tissue, modeled as a local increase of stiffness of the tissue. Analysis was performed on the movement of the breast without tumors and with the addition of a tumor of several sizes. This analysis found some measurable differences between the breast with tumor and without tumor, most notably in the x-displacement (side-to-side movement).



3 Methods

3.1 Breast geometry

One of the main goals of this project was to improve the geometry of the breast to better represent the asymmetric properties of the breast, and to further capture the variation in breast shape between individuals. In this section, the geometry of the unloaded (no gravity) breast is described.

Asymmetric shape of the breast In the previous model, the breast shape is defined as a quarter circle arc which is revolved around the y-axis. In this model the breast surface is shaped using a b-spline with control points on circle arcs with variable radii, forming a non-spherical shape as seen in Figure 4a. This radius is defined by a sequence of cosine functions :

$$r_v = r_b(1 + \sum_{i=1}^3 p_i(1 + \cos(i\theta))) \quad (6)$$

for $\theta \in [0, 2\pi]$ with input parameters breast radius r_b and asymmetry parameters p_{1-3} . By using $p_i(1 + \cos(i\theta))$, p_1 and p_3 are more intuitively scaled around 0, where a positive p_i results in an increased r_v for all θ , and a negative p_i results in a decrease in r_v for all θ .

Curved breast base In the previous model, the chest-side of the breast is a flat surface, while the breast in reality is situated on the curved ribcage area of the chest. This causes the breast to stand at an angle α from the front of the chest. In this model, this is simulated by building a sphere-like shape, and then cutting out the torso shape from the previous breast volume to achieve a half-sphere on a curved base. The curvature of the torso is approximated to a circle arc for simplicity, where the start of the slope of the breast in the negative x-direction is considered to be on the sternum. The radius of the torso shape is determined by

$$r_t = \frac{r_b}{2 \sin(\frac{\alpha}{2})} \quad (7)$$

such that the angle between the y-axis and the edge of the breast in the left half of the XY-plane is α (see Figure 4b). The torso radius is coupled to the nipple angle to simulate the position of the breast on the torso. If the angle is large, the breast is likely situated more on the side of the chest, which has higher curvature and is modeled as a smaller torso radius.

Adipose and glandular tissue In the model the glandular tissue is modeled as a scaled down copy of the breast volume, resulting in a layer of adipose tissue between the glandular tissue and the skin. There is little data on the thickness of the subcutaneous adipose layer, with large inter-subject variation, as shown in MRI imaging by Zhang et al. [17]. In the model the adipose layer is defined by scaling factors s_{xz} and s_y , the scaling factors in the x- and z- direction of the breast (towards the sides), and in the y-direction (towards the nipple), as pictured in Figure 4b

3.2 Meshing

The geometry is meshed by Gmsh's meshing algorithm, with base element size lc (default = 0.005 m). However, for areas with large deformations or shears, a smaller element size is desirable. Therefore, a variable mesh size is set to decrease the element size for the lower and upper regions of the breast, where most stretching and bending is expected to happen.

$$\text{mesh size} = (lc_{max} - lc_{min}) \cdot (\cos(z/z_{max} \cdot \frac{1}{2}\pi)) + lc_{min} \quad (8)$$

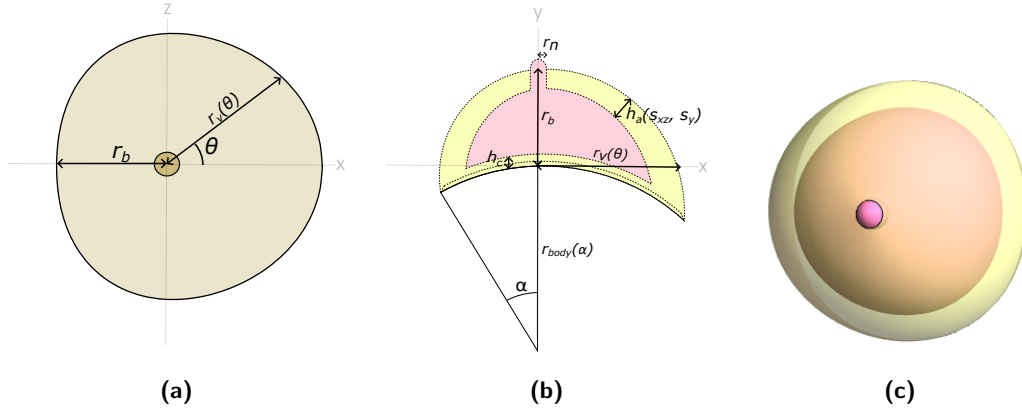


Figure 4: Construction of the breast model with skin (tan), adipose tissue (yellow) and glandular tissue (pink). **a)** Front view of the breast with varying radius r_v . **b)** Top view of the breast. **c)** 3D model of undeformed breast geometry with default parameters.

Parameter	Full name	Lower bound	Upper bound	Default
r_b (m)	Radius breast	0.05	0.15	0.07
r_n (m)	Radius nipple and duct	0.0075	0.002	0.0075
p_1 (-)	Asymmetry parameter 1	-1	1	0.1
p_2 (-)	Asymmetry parameter 2	-1	1	0.01
p_3 (-)	Asymmetry parameter 3	-1	1	0.01
α (deg)	Nipple angle	0	45	22.5
s_{xz} (-)	Scaling factor xz-direction	0.1	0.9	0.8
s_y (-)	Scaling factor y-direction	0.1	0.9	0.8

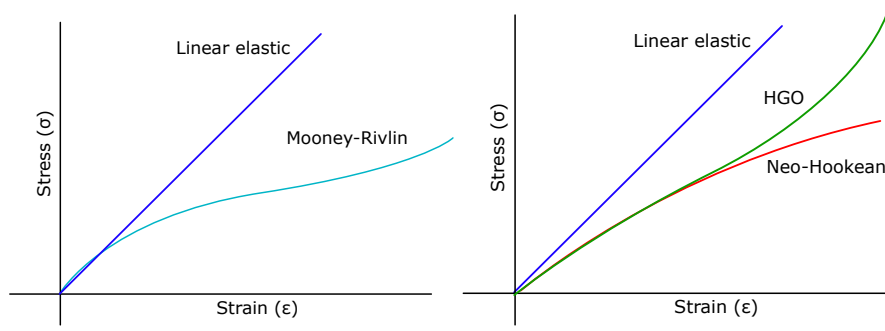
Table 1: Geometry parameters for variable breast geometry

Furthermore, the standard meshing by Gmsh creates elements on the edges of the model that have surfaces in both skin and chest surfaces. This causes shear locking, resulting in difficulties while running the FEM simulations. Therefore, a layer with thickness of ~ 1 element is segmented at the chest surface, which separates the model edge into two diagonally, to force the meshing of two elements at the edge (Fig 4b).

3.3 Finite Element Model

The breast is simulated using a Finite Element Method (FEM) model. Fem is a numerical method which provides approximate solutions for differential equation of a continuum by dividing it into small subdomains, also called finite elements. The differential equations are then approximated for these elements, which are then used to form an approximation for the global solution. Further background on FEM can be found in section 2.2.

For the solving of FEM problems, various commercial and non-commercial software packages are available. In this project FEM solver FEBio [18] was used. FEBio is a free and open-source solver which is focused on biological materials. The mesh was created using Gmsh's [19] Python API.



3.3.1 Material models

For FEM models, each material needs to be defined as a set of equations. For each tissue an existing material definition is used, where only the material parameters need to be selected based on the tissues mechanical properties.

Mooney-Rivlin In the previous model, the Mooney-Rivlin material model was selected for the tissues because of its frequent use in other models for describing the nonlinear properties of breast tissue. Mooney-Rivlin is a hyperelastic, isotropic material model. In FEBio, only the basic Mooney-Rivlin model is available, where the model's strain-energy function is given by:

$$\Psi_{MR} = C_1(\tilde{I}_1 - 3) + C_2(\tilde{I}_2 - 3) + \text{volumetric part} \quad (9)$$

with $\tilde{I}_1$ and $\tilde{I}_2$ the first and second invariants of the deviatoric right Cauchy-Green deformation tensor $\tilde{\mathbf{C}}$. The model coefficients C_1 and C_2 can be approximated from the initial shear modulus by

$$G_0 = 2(C_1 + C_2) \quad (10)$$

C_1 controls the primary shear and uniaxial stiffness, while C_2 controls large-strain deformations. The share of both coefficients is assumed to be roughly equal.

Holzapfel-Gasser-Ogden The Holzapfel-Gasser-Ogden material model describes a hyperelastic, fiber-reinforced material, resulting in anisotropic behavior of the material. This can be used to model the effect of collagen structures in the model, such as the suspensory ligaments.

The model consists of an isotropic ground matrix behaving as a Neo-Hookean material with two families of fibers in two directions. The strain-energy function is given by

$$\Psi_{HGO} = \frac{c}{2}(\tilde{I}_1 - 3) + \frac{k_1}{2k_2} \sum_{\alpha=1}^2 \left(\exp(k_2 \langle \tilde{E}_\alpha \rangle^2) - 1 \right) + U(J) \quad (11)$$

with fiber strain

$$\tilde{E}_\alpha = \kappa(\tilde{I}_1 - 3) + (1 - 3\kappa)(\tilde{I}_{4\alpha} - 1) \quad (12)$$

for fiber families $\alpha = 1, 2$. The ground matrix has shear modulus c , and the fibers are characterized by fiber modulus k_1 and exponential coefficient k_2 . The direction of the fibers is given by a vector $\mathbf{a}$, and fiber dispersion κ and angle γ , which describes the angle the two families have with respect to $\mathbf{a}$. If we set both families to lie in the direction of $\mathbf{a}$ by setting κ and γ to 0, we get

$$\Psi_{HGO} = \frac{c}{2}(\tilde{I}_1 - 3) + \frac{k_1}{k_2} \left(\exp(k_2 \langle \tilde{E} \rangle^2) - 1 \right) + \text{volumetric part} \quad (13)$$

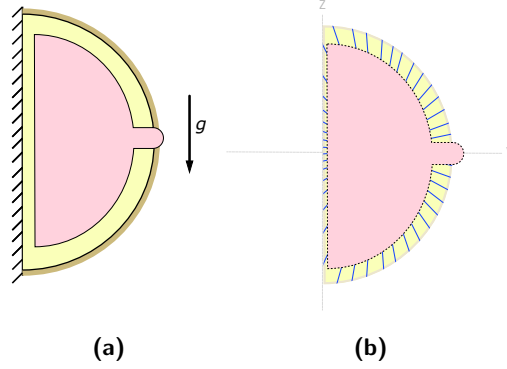


Figure 6: **a)** Side through-section of the breast model, with boundary condition on the chest surface and gravitational load g . **b)** Side through-section of the breast model with fibers in adipose tissue directed towards breast center)

3.3.2 Material parameters

Glandular tissue The glandular tissue is modeled with the Mooney-Rivlin material model, as the glandular tissue is assumed to be isotropic. This material was also used in the previous version of the model, where the shear modulus G was decided to be 0.85 kPa and C_1 and C_2 were chosen to be 230 and 195 Pa.

Adipose tissue and suspensory ligaments In the previous model, the adipose tissue was also simulated using the Mooney-Rivlin model, with shear modulus of 0.48 kPa and C_1 and C_2 as 106 and 109 Pa. One objective of this project was the addition of the suspensory ligaments in the adipose tissue. The suspensory ligaments were modeled using the Holzapfel-Gasser-Ogden material [20], where we assume a matrix with shear modulus of 0.48 kPa, and reinforcement with collagen fibers. For the collagen fibers fiber modulus k_1 and fiber exponential coefficient k_2 are set to be 5890 Pa and 12.5. Briot et al. [21] is the only source that was found to report values for the stiffness of suspensory ligaments in the breast, but this paper assumes a shear modulus of 0 for the ground matrix, and produce a very stiff breast. Alternatively, the values from the model in Holzapfel and Ogden [22] of the arterial wall are used. The fiber direction vector $\mathbf{a}$ is set by FEBio's built-in *spherical* direction option, where all fibers are oriented towards a set center point. Since the exact direction and structure of the suspensory ligaments is not conclusive, three different options are implemented. The first option is placing the center at the front of the breast near the nipple $(0, r_b, 0)$ to simulate the structure in Fig 2a. The second is to place the center in the center of the breast $(0, \frac{r_b}{2}, 0)$ to simulate the structure in Fig 2b, and lastly to add dispersion ($\kappa = 1/6$) to the fiber orientation to also simulate the perpendicular ligaments as in Fig 2c.

Skin The skin is modeled as hyperelastic isotropic shell elements of the Mooney-Rivlin material model, with thickness of 0.1 mm. This thickness is much lower than reported values for the thickness of breast skin [23]. However, modeling thicker skin brought up difficulties in the FEM model, so the layer of 0.1 mm can be seen as only modeling the stiffest part of the skin, the epidermis, where 0.1 mm falls within the observed range of thickness [24]

Glandular Tissue Mooney-Rivlin		Adipose Tissue Holzapfel-Gasser-Ogden		Skin Tissue Mooney-Rivlin		Tumor Tissue Mooney-Rivlin	
C_1	230 Pa	k_1	5890 kPa	C_1	1200 Pa	C_1	1200 Pa
C_2	195 Pa	k_2	12.5 kPa	C_2	1200 Pa	C_2	1200 Pa
ρ	1041 kg/m ²	γ	0	ρ	1100 kg/m ²	ρ	1100 kg/m ²
K	425000 Pa	κ	0	K	480000 Pa	K	480000 Pa
		ρ	1041 kg/m ²				
		K	425000 Pa				

Table 2: Material parameters for FEM model

Tumor In case a tumor is modeled, it is assigned the Mooney-Rivlin material model. Shear modulus is taken as $G_{tumor} = 0.43$ kPa, therefore parameters C_1 and C_2 are chosen to be 1080 Pa and 1045 Pa respectively.

3.3.3 Model parts

The FEM model is split up into a number of volumes and assigned surfaces. The main parts are the glandular and adipose tissues, where the glandular part consists of the main glandular tissue, the gland ducts and the nipple. The adipose part consists of the surrounding volume. If the model contains a tumor, this is added as an additional part. An issue in the model is that the tumor part can also include skin shell elements, if the tumor center is chosen too close to the skin.

Furthermore, we define the surface of the adipose tissue on the positive y-direction as the skin part, excluding the surface of the nipple. Lastly, the adipose surface in the negative y-direction is assigned as the chest part, which is required for the boundary condition.

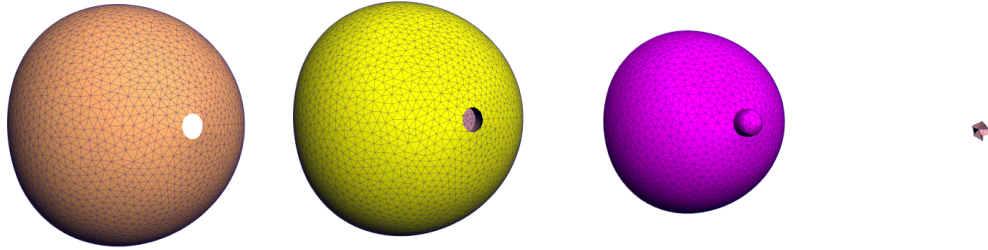


Figure 7: Parts of the breast model: Skin (orange), adipose (yellow), glandular (pink), tumor (gray)

3.3.4 Boundary conditions and loads

In this project the main focus is on the static image output of the model, and the dynamic movement of the breast is not taken into account to compare the model output to the available static data and reduce complexity. Therefore, only the boundary conditions and loads of the static step are considered. They are shown in Figure 6a.

Boundary conditions The breast has a zero displacement boundary condition in all dimensions at the breast base to simulate the attachment to the body. For this boundary condition the assumption

is made that there is no moving tissue posterior to the adipose chest wall. In reality, this wall is attached to the large chest muscles, which are not fully fixed. However, this movement is assumed to be negligible.

Loads To obtain the final output of the breast under gravitational load of, a load in the negative z-direction is increased from 0 to 9.81 m/s^2 over the duration of 1 second. This load is maintained for any further steps after 1 second.

3.4 Comparison of the model with real breast shapes

The breast created by the model should be able to accurately capture the variation of breast shapes within the population. Therefore, an algorithm is developed to match the model to a 3D scan of a breast to investigate whether the developed geometry parameters suffice to capture the geometry of real breasts.

3.4.1 Available data

The available data consists of 19 scans of the upper torso supplied by Fontys Hogescholen. The scans are supplied as .obj files. There are a few difficulties to the data. Firstly, the skin had a very high level of noise which translates so seemingly bumpy skin. Secondly, there are some gaps in the surfaces, often at the bottom of the breast. Some scans only capture the breast region, while others also include large parts of the arms, neck, and belly. Lastly, the scale of the scans in the files are not accurate, and are roughly 5 times too large.

Preparation of the data The .obj file is imported into python using the PyVista library [25]. It is first scaled to 0.2 of the original size, such that the breasts fall within realistic size. However, this scaling is an estimation, and accurate scaling should ideally be obtained from the scanner.

3.4.2 Matching algorithm

We have a scanned breast surface, built from a point cloud S_t , and the model-generated breast surface from point cloud S_m , where the points are the nodes from the FEM model. The two surfaces need to be aligned, and a measure distance between the two surfaces needs to be determined for optimizing the model geometry.

Alignment feature selection The first step is to find a feature to use for alignment of the surfaces of the scanned breast and the model breast. The breast has very few distinguishable features that can be used for alignment, with the nipple being the most obvious feature. Therefore, the point of the nipple is chosen as the alignment feature. For the target surface, the nipple is manually selected (Figure 8.1). Automatic selection may be favorable in the future, but due to the bumpy surface this would currently be difficult and unreliable. The surface mesh is translated such that the tip of the nipple lies at point (0,0,0).

The nipple is a useful alignment feature in this case, as in the unloaded model geometry the nipple always lies on the y-axis. Therefore, the target surface is rotated in such a way that the nipple points forward. To get a vector which is approximately in the direction of the nipple, the average normal for a region of 0.02m around the nipple is computed. The mesh is then rotated such that the nipple normal lies in the Y-Z plane (Figure 8.2, 8.3). The model surface is also translated such that the tip of the nipple is in the origin. However, the nipple does not need to be manually set, but it is instead determined by finding the surface point with the largest y-coordinate in the mesh.

Segmentation of target breast The target breast is segmented from the rest of the scan. The model only produces one synthetic breast, so one breast is segmented from the scan surface. This is done manually, by selecting a number of points around the breast which are connected by a spline. All surface points within the projection of this spline in the y-direction are included in the segmented breast surface (Figure 8.4, 8.5).

Distance between surfaces Both surfaces are now oriented the same (Figure 8.6), so a measure of distance can be computed. For each point p in S_m , the closest point q^* in S_t can be found using SciPy KDTree [26]. This function can now be used by the optimization algorithm.

3.4.3 Optimization

For the optimization, the derivative-free optimizer LIMOLS is used. This type of optimizer is useful for problems where each iteration is computationally expensive. In this case, a single iteration can take from 2 to up to 20 minutes, making it essential to limit the number of required iterations.

The objective is to determine the set of geometric parameter values which minimizes the distance between the model surface and the target surface. LIMOLS requires a fixed number of residuals as input and performs better with multiple residuals as input. Therefore, a fixed number of points is defined for which the surface distance is computed. As the size of the model breast changes every iteration, 200 points are projected on the model surface in uniformly distributed lines from the origin (Figure 8.7), giving $p_i \in S_m$ for $i = [0, 200]$. The input for LIMOLS is the difference $p_i - q_i$ in three dimensions for all p_i , resulting in a vector of length 600: $(x_{p_1} - x_{q_1}, y_{p_1} - y_{q_1}, \dots, z_{p_{200}} - z_{q_{200}})$. It is important not to take the absolute values of the distances, as LIMOLS performs best with residuals that can be both positive and negative values.

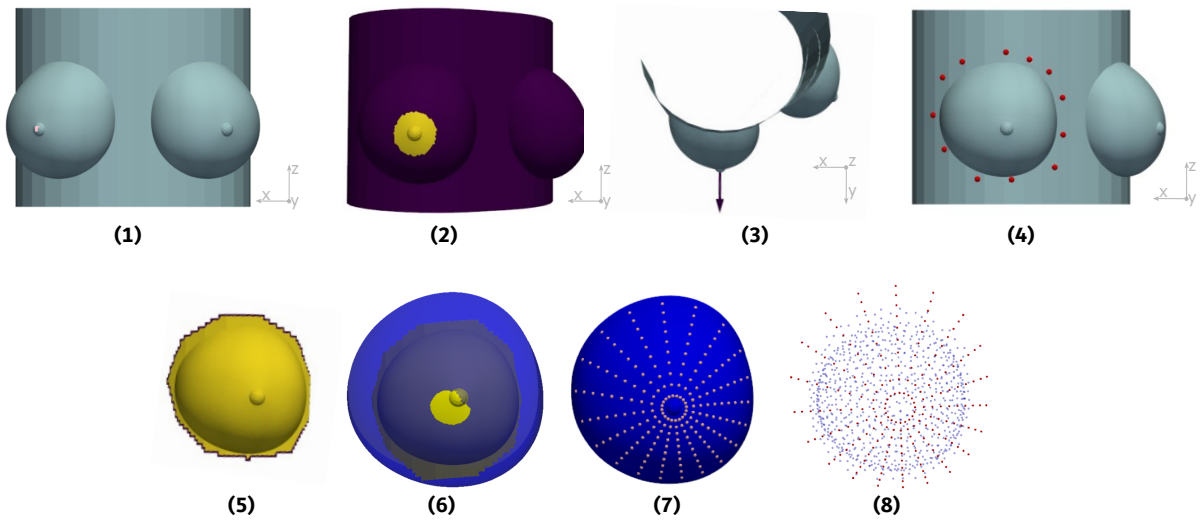


Figure 8: Optimization procedure: **1)** Imported surface with selected nipple point. **2)** Selected nipple area for nipple direction. **3)** Rotated surface with nipple normal. **4)** Selection of target breast area. **5)** Segmented target breast. **6)** Target breast (yellow) with aligned initial model breast (blue) at iteration 0. **7)** Projected points on model breast. **8)** Projection points with point cloud of target breast.

4 Results

4.1 Variable geometry

As described in section 3.1, the geometry of the breast was parametrized using 5 external parameters and 3 internal geometric parameters. Default geometric parameters can be found in Table 1, and the resulting geometry deformed under gravity can be found in Figure 9.

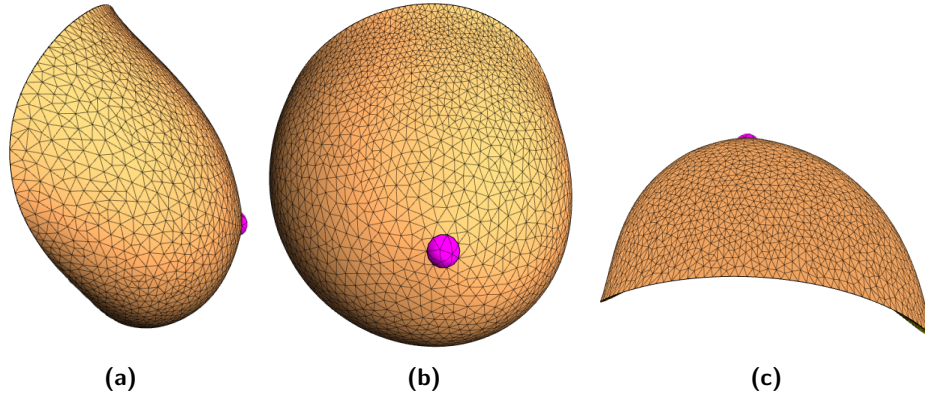


Figure 9: a-c) Side, front, and top view of breast with all default parameter values with Mooney-Rivlin material for all tissues.

The default breast shape is slightly asymmetric, with more volume on the left side of the nipple, similar to what can be seen in the left breast in breast (MRI-) images. In Figure 10, some examples are shown of what effect the asymmetry parameters have on the breast. p_1 determines the size of the left side of the breast compared to the right side (larger for positive values and smaller for negative values). p_2 tunes the flatness of the breast, with positive values resulting in a more vertically flat breast (see Figure 10c) and negative values in a more horizontally flat breast (see Figure 10d). p_3 adds a slight triangular shape to the breast, as can be seen in Figure 10b.

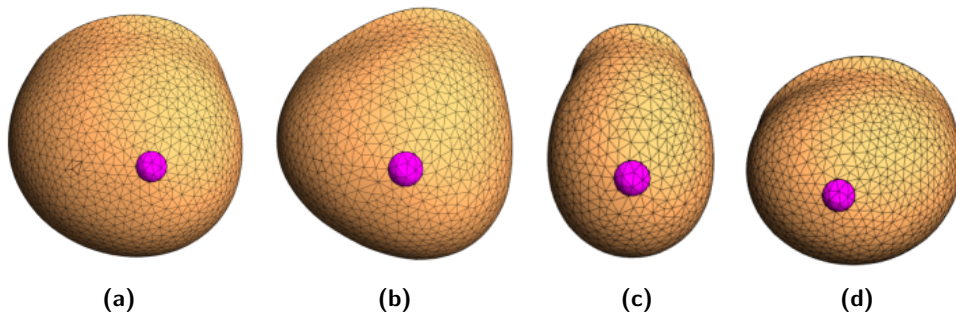


Figure 10: Some examples of the breast model with identical radii (0.05 m) and varying values for p_{1-3} . a) Default values. b) $p_3=0.1$, $p_2=0.03$. c) $p_2 = -0.15$. d) $p_1 = -0.15$, $p_2 = 0.1$

4.2 Suspensory ligaments

For the suspensory ligaments, the Holzapfel-Gasser-Ogden model was implemented with fibers in the direction of a set center point as described in section 3.3.2.

In Figure 11 the resulting breast meshes are shown. Figure 11a, the model with no suspensory ligaments, shows strong resemblance to the model using the Mooney-Rivlin material model for the adipose tissue, such as the mesh in Figure 9a.

In Figure 11b the breast with fibers towards the center of the breast is shown. The addition of fibers causes a visible difference in the shape of the top of the breast, where there is less displacement. To further investigate this change, the internal strains for the breast is revealed in Figure 12. It becomes apparent that the strain in the breast tissue is generally highest near the boundary of the chest wall. In the upper half of the chest wall, the fibers are directed in the direction of the gravitational load in the model, and therefore the fibers resist strain the most in this area. This lowered strain in the chest wall causes the bump that is visible on the top of the breast. The consequent reduction in lowering. The addition of dispersion of the fibers does not cause a large difference in the final shape of the breast. The addition of dispersion to the fiber direction retains this effect, as shown in Figure 11c, but does not cause any further major visible changes to the breast shape.

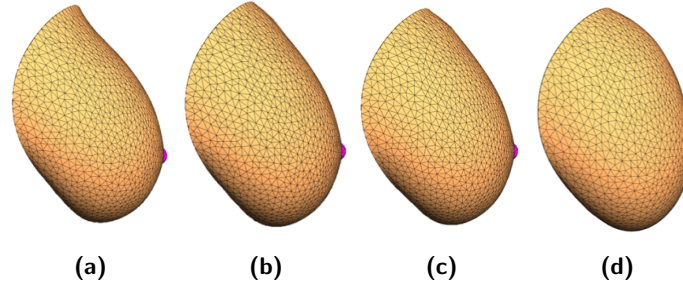


Figure 11: Breast model output with HGO material for adipose tissue. **a)** Without fibers ($k_1 = 0$). **b)** Fibers towards the breast center. **c)** Fibers towards center of breast, with dispersion. ($\kappa = 1/6$) **d)** Fiber towards nipple.

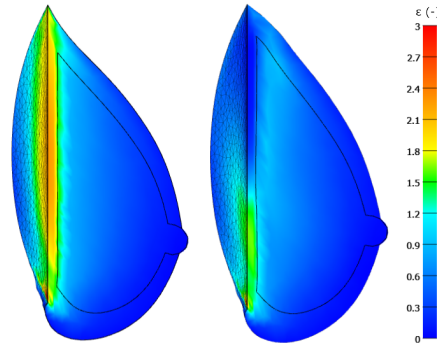


Figure 12: Strain throughout the breast for breast without fibers (left) and with fibers towards center (right).

4.3 Parameter optimization

An algorithm was developed to optimize the geometric parameters to generate a breast with the closest distance to an input target surface. As a sanity check, the optimization was tested on a synthetic breast surface with known geometry parameters. In this optimization task, parameters r_b and α_n were optimized to a synthetic breast surface with known $r_b = 0.048$ and $\alpha_n = 31.7$, and default values for all other (as found in Table 1 and 2). The optimized parameter values from the algorithm (after a maximum of 20 iterations) were 0.048 and 34.2 with a residual norm of 0.011.

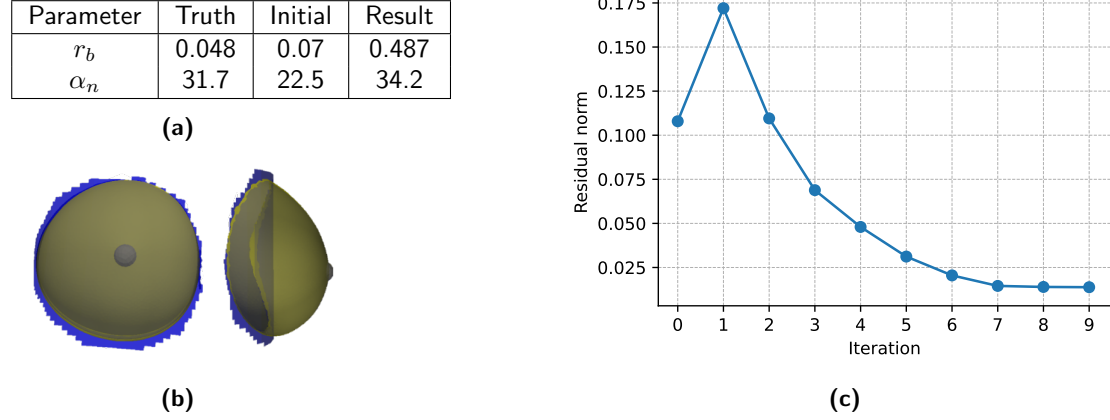


Figure 13

For the optimization to the dataset of 3D breast scans the optimization was first performed for r_b and α_n . The quality of the results vary widely per individual case. The exact result cannot be shown here, but approximate side profiles for three cases are shown in Figure 14. In Figure 14a the case with the best optimization result is displayed, with an eventual residual norm of 0.025. For this case, the target breast is relatively round and stiff, resulting in a closer match with the model breast. In Figure 14b a case is shown where the target breast is much less round than the model, which results in a worse fit with the model breast and a residual norm of 0.054. In Figure 14c the target breast has a round bottom but a flat top, which results in a worse fit with the model breast and a residual norm of 0.054.

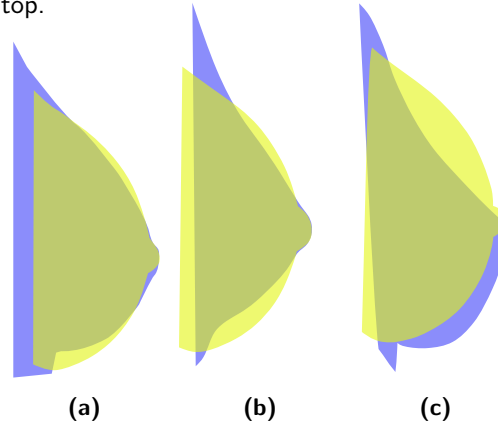


Figure 14: Side profile of optimization results with target (blue) and model breast (yellow).

Due to poor performance of this optimization with two parameters, only limited further optimization into other geometric parameters was performed, but without mentionable result. Optimization of stiffness parameters C_1 , C_2 , c , k_1 and k_2 was not feasible due to problems in the FEM simulations, which will be elaborated on in section 5.2.

5 Discussion

5.1 Breast geometry

The added asymmetry in the breast model adds realistic features to the breast, such as more breast volume in the region of the axillary tail. Very little research is available on the exact shape of the breast, and in most similar models the breast shape is taken from a real subject, instead of a parameterized geometry. This model is unique in its parameterized geometry, which is an advantage in its goal of creating large amounts of data.

One result of the methods used for the asymmetry is that some asymmetric features can only be formed on one side. The variable radius of the breast as presented in section 3.1 starts on the positive x -axis of the breast. Asymmetry parameters p_{1-3} form the trajectory of the radius, but the start point remains the same. Therefore, the model is unable to create mirrored shapes using the parameters. One solution may be to add a (binary) parameter which determines the phase of the variable radius calculation.

Additionally, in this project the focus of the variable breast shape was mostly focused on the front-view shape of the initial breast shape and the shape of the chest wall. However, in the breast shape optimization, some of the results suggest that the sphere-like shape of the breast does not always suffice. In Figure 14b and 14c we see a breast shape that is more pointed rather than round, which would not be possible with the current geometric model. In future iterations of the model, a measure of curvature of the breast could be introduced to also include these shapes.

5.2 Material parameters

The material parameters are taken from in-vivo and cadaver measurements from various literature sources. However, while already using material parameters on the low end of what literature provides, the resulting breast does not seem to recreate the shape of the breast that we see in real-life imaging of the breast. The main difference is that the breasts in a large portion of the data exhibit more ptosis (hanging of the breast) than the breast generated by the model. This could be caused both by material parameters and by initial geometry. However, in Mîra et al. [27] the geometry is taken from MRI imaging, and it is concluded that measurement values from literature do not suffice in recreating the softness of the breast, and better results are obtained by optimizing the material parameters. This affirms the idea that the problem may lie mostly in the parameter values. However, simply lowering the shear modulus of the breast tissues to decrease stiffness was not a possibility.

5.3 Element inversion

The deformation of the breast under gravity brings a large deformation to the elements in the mesh. This often causes inverted elements, resulting in a negative jacobian error in FEBio. FEBio's solution to this is to lower the timestep and reform the stiffness matrix. However, with the large deformation that we see in parts of the breast, this is not enough to stop the elements from inverting. Initially, the model encountered negative jacobians for nearly every simulation. A large part of this issue was solved by splitting the adipose domain to prevent mesh locking at the bottom of the breast, as described in section 3.2. However, the problem still persist when lowering the shear modulus of the material lower than ~ 300 Pa. With this low stiffness, the displacement becomes so large that FEBio cannot deal with it. In this case remeshing (defining a new mesh while and projecting the previous stresses, strains, pressures on the new mesh) would be a good solution, but FEBio does not have this option. In other models, such as the one in Mîra et al. [27] and Chen et al. [28], other (commercial) FEM

solvers are used, and these models do not suffer the limitations of stiffness.

5.4 Suspensory Ligaments

In this project, the suspensory ligaments are introduced to the model, to represent the shaping function they have for the breast. However, there is little consensus in literature on the exact structural and mechanical properties of these ligaments. Firstly, as mentioned in section 2.1.2, there are several theories on the position and orientation of the ligaments. In this model, we chose to model the suspensory ligaments implicitly through the use of the Holzapfel-Gasser-Ogden model. This choice was motivated by the difficulty of modeling thin structures in FEM, and the added complexity of explicitly modeling the ligaments. In other literature, the suspensory ligaments are often neglected [29, 30, 31], while Chen et al. [28] models the ligaments as several long, cylinder-like structures throughout the breast, which does not match descriptions of the ligament in dissection studies [14, 16, 32]. For the suspensory ligaments parameters k_1 and k_2 were set to values from Holzapfel and Ogden [22], who model the arterial wall. Values measured specifically for suspensory ligaments were available from Briot et al. [21], but these resulted in an extremely stiff breast. The values from Holzapfel and Ogden [22] are likely too high, as the fiber density in the arterial wall is much higher than in the breast, and more realistic values should be explored in further iterations of the model. However, this was beyond the scope of this project.

5.5 Parameter optimization

A parameter optimization pipeline was developed to match model output to real-life 3D scans, and to evaluate the match between the two surfaces. The initial performance on a breast with known parameters proved the optimization to be promising. The results of the parameter optimization to the dataset as seen in section 4.3 vary vastly between subjects. Some issues to do with geometry and material parameters were discussed in section 5.2 and 5.1 and can be responsible for poor performance in the optimization.

One challenge in the comparison of the model to the target breast is the alignment of the surfaces. There is no exact definition of the breast area, and therefore the target surface may include more than the area recreated by the model. For the alignment of the two surfaces, the only feature available on every breast is the nipple. In the model the nipple is always in the middle of the breast in the undeformed state. While this may not be true for every breast, we do see that it is relatively accurate for most breasts in the dataset. Furthermore, the model assumes the nipple to dictate the direction of the breast. While this alignment may not be fully accurate, the results of the optimization did not show clear problems due to the assumptions made, and the alignment method is simple and effective.

This alignment by the nipple does expose a major issue in the optimization process. Often, it is observed that the nipple on the model breast is situated higher than the nipple in the target breast. This is due to the lack of ptosis in the model, as a result of the high stiffness and inability to lower the stiffness as discussed in 5.2.

In the current optimization pipeline, residuals are calculated using 200 points projected onto the front of the breast, distributed evenly from center to periphery. This results in an uneven sampling of the full breast surface, with the bottom and sides underrepresented due to curvature. Distributing points over the entire surface introduces its own challenges, the region of the inframammary fold (where the bottom of the breast rests against the torso) are not visible in scans but are exposed in the model due to the absence of the torso. A possible improvement is to use a non-uniform point distribution with higher density near the periphery, or to project points toward the breast center.

In some cases the target breast surface may cause inaccuracies. The surface, acquired using the EWS scanning setup, is relatively noisy. There are many bumps in the surface, sometimes almost as large as the nipple. This noise will have some impact on the quality of the optimization, as the optimization uses the distance to the closest point on the surface. This will bias the optimization towards a slightly larger radius.

Furthermore, the target breast is approximately segmented from the surrounding skin surface through manual selection. This approach leaves some residual torso skin around the breast, particularly in the region beneath it. In some optimization cases, such as in Figure 14a the model breast overlaps with the torso under the breast, resulting in an incorrectly low residual for some projection points. However, this issue would be mostly resolved by addressing the stiffness issue.

In this project, only two parameters were optimized, but accurate representation of the full geometry will require optimization of a larger set of parameters, including other geometric parameters and possibly material parameters. With the current dataset, it is unlikely that a unique solution can be found for a large amount of parameters. It would be beneficial in future research to have more data available per subject, such as different positions (e.g. supine) or dynamic data.

5.6 Summary

This project introduced a parameterized geometry capable of representing asymmetric breast shapes, incorporated the Holzapfel–Gasser–Ogden material model to account for the effect of the suspensory ligaments, and established a pipeline for optimizing the model geometry against a dataset of 3D breast images. While the increased geometric flexibility and the optimization framework show promising initial results, the overall model performance is strongly constrained by limitations in material parameter selection, mostly due to numerical challenges. Future work should first focus on addressing these numerical limitations, after which further refinement of both geometric and material parameters can be done. These developments will enhance the realism of the generated data and contribute to the development of the EWS.

Bibliography

1. Bray F, Laversanne M, Sung H, Ferlay J, Siegel RL, Soerjomataram I, and Jemal A. Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. eng. CA: a cancer journal for clinicians 2024; 74:229–63. DOI: [10.3322/caac.21834](https://doi.org/10.3322/caac.21834)
2. World Health Organization. Breast cancer. en. 2025 Aug
3. Henry NL, Shah PD, Haider I, Freer PE, Jagsi R, and Sabel MS. Cancer of the Breast. en. *Abeloff's Clinical Oncology*. Elsevier, 2020 :1560–1603.e12. DOI: [10.1016/B978-0-323-47674-4.00088-8](https://doi.org/10.1016/B978-0-323-47674-4.00088-8)
4. WebMD. Breast Cancer Survival Rates. en. 2024 Aug
5. Zhang Y, Rodriguez J, Mao X, Grassmann F, Tapia J, Eriksson M, Hall P, and Czene K. Incidence and Risk Factors of Interval and Screen-Detected Breast Cancer. JAMA Oncology 2025 May; 11:519–27. DOI: [10.1001/jamaoncol.2025.0167](https://doi.org/10.1001/jamaoncol.2025.0167)
6. American Cancer Society. Cancer Facts & Figures 2021. en. Tech. rep. 2021
7. Gurbani SS, Ikuta I, Makary MS, Akram M, Kantamneni D, Azizaddini S, Judd D, Sotelo J, Lanser EM, Chetlen A, and Chhor C. Ionizing Radiation Exposure: What are the Risks Today? Academic Radiology 2025 Dec; 32:7012–22. DOI: [10.1016/j.acra.2025.08.030](https://doi.org/10.1016/j.acra.2025.08.030)
8. Geuzinge HA, Bakker MF, Heijnsdijk EAM, Ravesteyn NT van, Veldhuis WB, Pijnappel RM, Lange SV de, Emaus MJ, Mann RM, Monninkhof EM, Koekkoek-Doll PK de, Gils CH van, Koning HJ de, and DENISE trial study group. Cost-Effectiveness of Magnetic Resonance Imaging Screening for Women With Extremely Dense Breast Tissue. Journal of the National Cancer Institute 2021; 113:1476–83. DOI: [10.1093/jnci/djab119](https://doi.org/10.1093/jnci/djab119)
9. Ackerson K and Preston SD. A decision theory perspective on why women do or do not decide to have cancer screening: systematic review. en. Journal of Advanced Nursing 2009; 65. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1365-2648.2009.04981.x>:1130–40. DOI: [10.1111/j.1365-2648.2009.04981.x](https://doi.org/10.1111/j.1365-2648.2009.04981.x)
10. YvYa foundation. Early Warning Scan. nl
11. Bazira PJ. Anatomy and physiology of the breast. Surgery (Oxford). Breast 2024 Dec; 42:859–64. DOI: [10.1016/j.mpsur.2024.09.003](https://doi.org/10.1016/j.mpsur.2024.09.003)
12. Jesinger RA. Breast Anatomy for the Interventionalist. Techniques in Vascular and Interventional Radiology. Breast Interventions 2014 Mar; 17:3–9. DOI: [10.1053/j.tvir.2013.12.002](https://doi.org/10.1053/j.tvir.2013.12.002)
13. Wiskin J, Klock J, and Love S. Breast Glandular and Ductal Volume Changes during the Menstrual Cycle: A Study in 48 Breasts Using Ultralow-Frequency Transmitted Ultrasound Tomography/Volography. en. Tomography 2024 May; 10:789–805. DOI: [10.3390/tomography10050060](https://doi.org/10.3390/tomography10050060)

14. Rehnke RD, Groening RM, Van Buskirk ER, and Clarke JM. Anatomy of the Superficial Fascia System of the Breast: A Comprehensive Theory of Breast Fascial Anatomy. *Plastic and Reconstructive Surgery* 2018 Nov; 142:1135–44. DOI: [10.1097/PRS.0000000000004948](https://doi.org/10.1097/PRS.0000000000004948)
15. Gefen A and Dilmoney B. Mechanics of the normal woman's breast. *Technology and health care : official journal of the European Society for Engineering and Medicine* 2007; 15. 259:259–71
16. Gaskin KM, Peoples GE, and McGhee DE. The fibro-adipose structure of the female breast: A dissection study. en. *Clinical Anatomy* 2020; 33:146–55. DOI: [10.1002/ca.23505](https://doi.org/10.1002/ca.23505)
17. Zhang Y, Chen JH, Chang KT, Park VY, Kim MJ, Chan S, Chang P, Chow D, Luk A, Kwong T, and Su MY. Automatic Breast and Fibroglandular Tissue Segmentation in Breast MRI Using Deep Learning by a Fully-Convolutional Residual Neural Network U-Net. *Academic Radiology* 2019 Nov; 26:1526–35. DOI: [10.1016/j.acra.2019.01.012](https://doi.org/10.1016/j.acra.2019.01.012)
18. Weiss Biomechanics Lab and Musculoskeletal Biomechanics Laboratory Columbia. FEBio Software Suite
19. Geuzaine C and Remacle JF. Gmsh: a three-dimensional finite element mesh generator with built-in pre- and post-processing facilities. en. *International Journal for Numerical Methods in Engineering* 2009
20. Gasser TC, Ogden RW, and Holzapfel GA. Hyperelastic modelling of arterial layers with distributed collagen fibre orientations. *Journal of The Royal Society Interface* 2005 Sep; 3:15–35. DOI: [10.1098/rsif.2005.0073](https://doi.org/10.1098/rsif.2005.0073)
21. Briot N, Chagnon G, Burlet L, Gil H, Girard E, and Payan Y. Experimental characterisation and modelling of breast Cooper's ligaments. *Biomechanics and Modeling in Mechanobiology* 2022; 21:1157–68. DOI: [10.1007/s10237-022-01582-5](https://doi.org/10.1007/s10237-022-01582-5)
22. Holzapfel GA and Ogden RW. Modeling the biomechanical properties of soft biological tissues: Constitutive theories. *European Journal of Mechanics - A/Solids* 2025 Jul; 112:105634. DOI: [10.1016/j.euromechsol.2025.105634](https://doi.org/10.1016/j.euromechsol.2025.105634)
23. Huang SY, Boone JM, Yang K, Kwan ALC, and Packard NJ. The effect of skin thickness determined using breast CT on mammographic dosimetry. eng. *Medical Physics* 2008 Apr; 35:1199–206. DOI: [10.1118/1.2841938](https://doi.org/10.1118/1.2841938)
24. Epidermal thickness in healthy humans: a systematic review and meta-analysis - Lintzeri - 2022 - *Journal of the European Academy of Dermatology and Venereology* - Wiley Online Library
25. Sullivan CB and Kaszynski AA. PyVista: 3D plotting and mesh analysis through a streamlined interface for the Visualization Toolkit (VTK). en. *Journal of Open Source Software* 2019 May; 4:1450. DOI: [10.21105/joss.01450](https://doi.org/10.21105/joss.01450)
26. Maneewongvatana S and Mount D. Analysis of approximate nearest neighbor searching with clustered point sets. 2002 Dec :105–23. DOI: [10.1090/dimacs/059/06](https://doi.org/10.1090/dimacs/059/06)
27. Mîra A, Carton AK, Muller S, and Payan Y. A biomechanical breast model evaluated with respect to MRI data collected in three different positions. en. *Clinical Biomechanics* 2018 Dec; 60:191–9. DOI: [10.1016/j.clinbiomech.2018.10.020](https://doi.org/10.1016/j.clinbiomech.2018.10.020)
28. Chen J, Zhong Z, Sun Y, Yip J, and Yick KI. Dynamic simulation of breast behaviour during different activities based on finite element modelling of multiple components of breast. en. *Scientific Reports* 2025 Jan; 15:3659. DOI: [10.1038/s41598-024-83598-8](https://doi.org/10.1038/s41598-024-83598-8)
29. Cai Y, Chen L, Yu W, Zhou J, Wan F, Suh M, and Chow DHk. A piecewise mass-spring-damper model of the human breast. *Journal of Biomechanics* 2018 Jan; 67:137–43. DOI: [10.1016/j.jbiomech.2017.11.027](https://doi.org/10.1016/j.jbiomech.2017.11.027)

30. Chen LH, Ng SP, Yu W, Zhou J, and Wan KWF. A study of breast motion using non-linear dynamic FE analysis. *Ergonomics* 2013 May; 56:868–78. DOI: [10.1080/00140139.2013.777798](https://doi.org/10.1080/00140139.2013.777798)
31. Zhang J, Lau NM, Sun Y, Yip J, Yick KL, Yu W, and Chen J. Non-linear finite element model established on pectoralis major muscle to investigate large breast motions of senior women for bra design. *EN. Textile Research Journal* 2022 Oct; 92:3511–21. DOI: [10.1177/00405175221075049](https://doi.org/10.1177/00405175221075049)
32. Cardoso A, Santos D, Martins J, Coelho G, Barroso L, and Costa H. Breast ligaments: an anatomical study. 2014 Sep; 38:91–6. DOI: [DOI10.1007/s00238-014-1024-7](https://doi.org/10.1007/s00238-014-1024-7)